HOME LAB NETWORK
DOCUMENTATION MANUAL

Version 2.0 — Post Phase 2 (Complete)

Protectli FW6E • pfSense 2.8.1 • Mullvad VPN (WireGuard) • Dual-Tunnel Failover
Netgear GS308E v4 • Netgear R6400 AP • pfBlockerNG • Tailscale

PERSONAL USE ONLY — CONTAINS SENSITIVE CONFIGURATION DATA (Public ver.)

Last Updated: March 02, 2026

Table of Contents

1. Document Control

This document tracks the live configuration of the home lab network. All changes are logged with date, description, and backups taken.

1.1 Document Information

Field	Value
Document Title	Home Lab Network Documentation Manual v2.0
Classification	Personal / Confidential — Do Not Distribute
pfSense Version	2.8.1-RELEASE (amd64) — FreeBSD 15.0-CURRENT
Hostname / Domain	mother.mother.net
Document Version	2.0 — Post Phase 2 (all VLANs, dual VPN failover, full firewall)

1.2 Change Log

Date	Section	Change	Backup
Mar 02, 2026	All	Initial documentation v1.0	pfsense-initial.xml
Mar 02, 2026	Kill Switch	Fixed 5 issues: IPv6 leak, DNS lockdown, resolver restriction	after-killswitch-audit.xml
Mar 02, 2026	Interfaces	Renamed: VPN_CHI, VLAN10_USERS, RESERVED_OOB/LAB/HA/EXP	before-phase2.xml
Mar 02, 2026	VLANs	Created 20(IOT), 30(GUEST), 40(LAB), 50(MGMT)	before-lan-renumber.xml
Mar 02, 2026	Subnets	LAN→10.10.1.0/24, VLAN10→10.10.10.0/24	before-vlan10-renumber.xml
Mar 02, 2026	VPN	Added NYC tunnel, VPN_FAILOVER group (CHI=T1, NYC=T2)	
Mar 02, 2026	Firewall	Full rules: 7 interfaces, RFC1918, DOH, DNS blocks	
Mar 02, 2026	Switch	802.1Q: 6 VLANs, trunk P1, access P2-8	switch config exported
Mar 02, 2026	AP	R6400 IP updated to 10.10.10.254, AP mode confirmed	
Mar 02, 2026	Testing	All passed: VPN, DNS, isolation, failover, kill switch	final-backup.xml

☑ **HINT** Update this log EVERY time you change the network. First place to check when troubleshooting.
☑ **NOTE** Mar 02, 2026 — last log update (lab in progress for over 7 months)

2. Hardware & Software Inventory

2.1 Hardware Assets

Device	Model / Specs	Role	Management IP
Firewall	Protectli FW6E — i7-8550U, 16GB DDR4, 6x 1GbE	pfSense firewall, VPN, DHCP, DNS	10.10.1.1 / 10.10.10.1
Switch	Netgear GS308E v4 — 8-port GbE (S/N: xxxxxxxxxxx8A)	802.1Q VLAN trunking	10.10.1.100
AP	Netgear R6400 AC1750 — AP mode only	Wi-Fi for VLAN 10	10.10.10.254
Lab Switch	Cisco Catalyst 3560	CCNA/CCNP lab	N/A
Lab Router	Cisco 1900 Series	CCNA/CCNP lab	N/A

2.2 Software

Software	Version	Purpose
pfSense CE	2.8.1-RELEASE	Firewall OS (FreeBSD 15.0)
WireGuard	Latest	VPN protocol — 2 tunnels (CHI 51820, NYC 51821)
pfBlockerNG	3.2.8	DNS ad/threat blocking (VIP: 10.10.255.1)
Tailscale	Installed	Remote mesh VPN access
Mullvad VPN	WireGuard	Privacy VPN — Chicago + NYC

3. Network Design

Router-on-a-stick: all VLANs on single 802.1Q trunk (igb1). All internet exits through Mullvad VPN with auto-failover. 5-layer kill switch prevents ISP leaks.

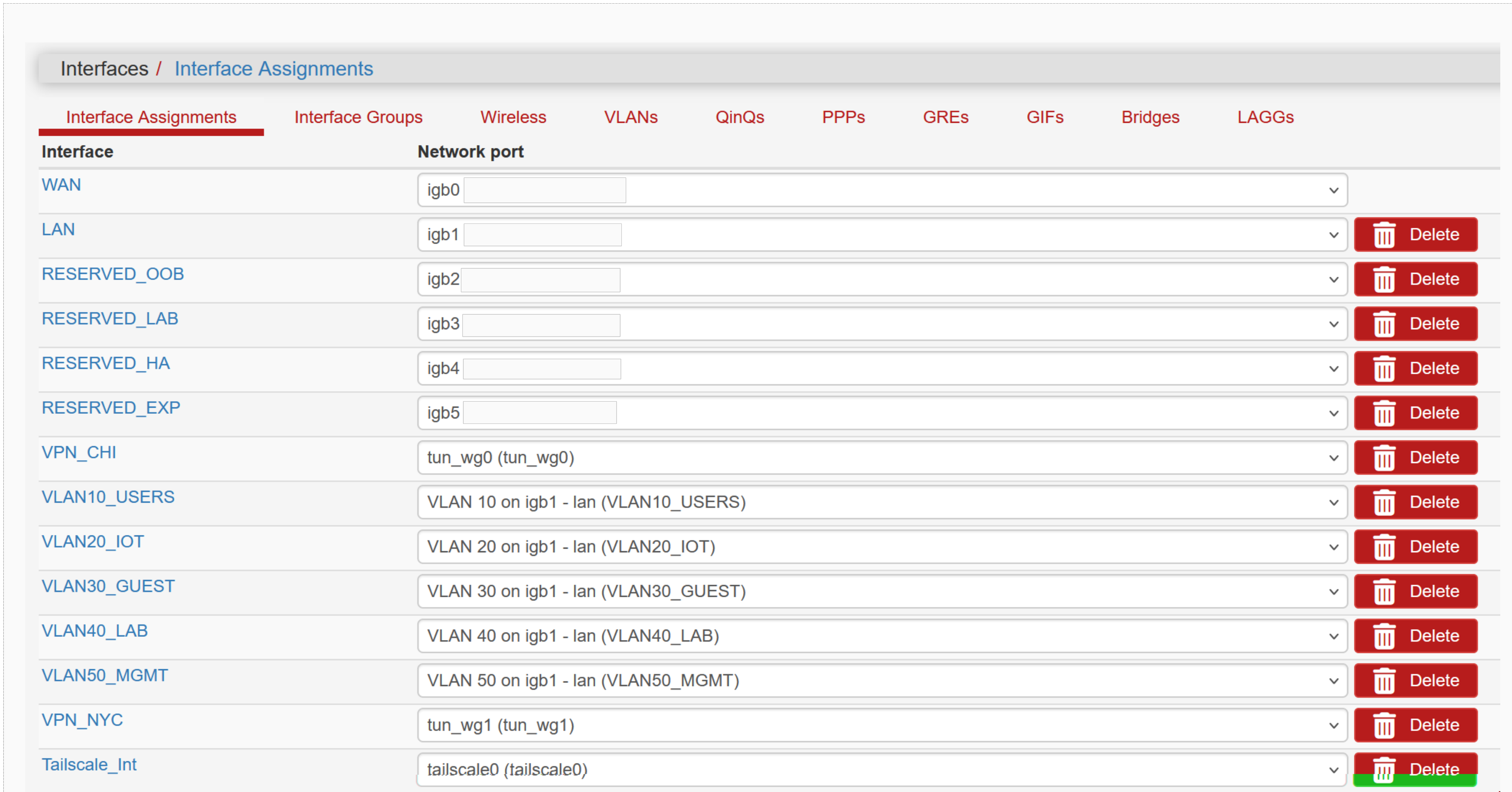
3.1 VLAN Summary

VLAN	Name	Subnet	Gateway	VPN	Purpose
1	LAN (Native)	10.10.1.0/24	10.10.1.1	Yes	Infrastructure / switch mgmt
10	VLAN10_USERS	10.10.10.0/24	10.10.10.1	Yes	PCs, phones, Wi-Fi
20	VLAN20_IOT	10.10.20.0/24	10.10.20.1	Yes	Printer, IoT devices
30	VLAN30_GUEST	10.10.30.0/24	10.10.30.1	Yes	Guest Wi-Fi (future)
40	VLAN40_LAB	10.10.40.0/24	10.10.40.1	Yes	Cisco lab gear
50	VLAN50_MGMT	10.10.50.0/24	10.10.50.1	No	Break-glass management

INFO Subnet = VLAN ID in 3rd octet (VLAN 10 = 10.10.10.x). Any engineer identifies the VLAN from the IP instantly.

3.2 Interface Assignments

Interface	Port	IP	Status	Description
WAN	igb0	DHCP (ISP)	Active	ISP uplink
LAN	igb1	10.10.1.1/24	Active	Trunk + native VLAN 1
RESERVED_OOB	igb2	None	Disabled	Emergency OOB
RESERVED_LAB	igb3	None	Disabled	Cisco lab direct
RESERVED_HA	igb4	None	Disabled	Future HA/pfSync
RESERVED_EXP	igb5	None	Disabled	Future expansion
VPN_CHI	tun_wg0	Mullvad	Active	WireGuard — Chicago
VPN_NYC	tun_wg1	Mullvad	Active	WireGuard — New York
VLAN10_USERS	igb1.10	10.10.10.1/24	Active	User devices
VLAN20_IOT	igb1.20	10.10.20.1/24	Active	IoT / Printer
VLAN30_GUEST	igb1.30	10.10.30.1/24	Active	Guest
VLAN40_LAB	igb1.40	10.10.40.1/24	Active	Lab
VLAN50_MGMT	igb1.50	10.10.50.1/24	Active	Management



pfSense Dashboard + Interfaces > Assignments

4. VPN — Mullvad WireGuard + Failover

Dual-tunnel: Chicago primary (Tier 1), New York failover (Tier 2). Gateway group VPN_FAILOVER handles automatic switching.

4.1 Tunnels

	Chicago (Primary)	New York (Failover)
Tunnel	TUN_WG_CHI	TUN_WG_NYC
Interface	VPN_CHI (tun_wg0)	VPN_NYC (tun_wg1)
Port	51820	51821
Gateway	GW_VPN_CHI	GW_VPN_NYC
Monitor	1.1.1.1	9.9.9.9
Traffic	IPv4 only	IPv4 only

4.2 Gateway Group

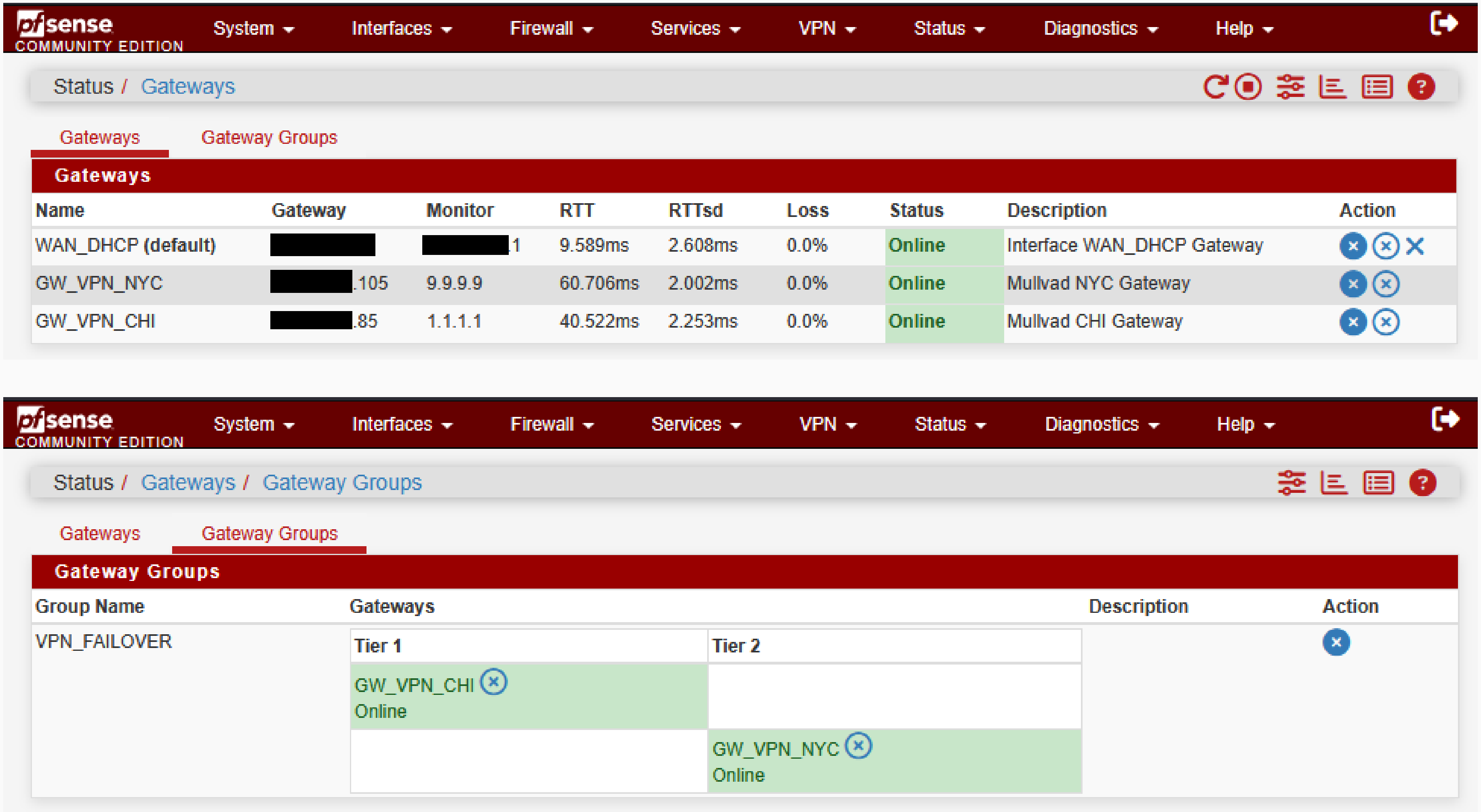
Name	Tier 1	Tier 2	Trigger
VPN_FAILOVER	GW_VPN_CHI	GW_VPN_NYC	Member Down

4.3 DNS Split

DNS Server	Gateway	Why
100.64.0.1	GW_VPN_CHI	Primary DNS through primary tunnel
100.64.0.2	GW_VPN_NYC	Backup DNS through failover tunnel

4.4 Kill Switch (5 Layers)

Layer	What	Where	If Missing
1. Routing	All rules use VPN_FAILOVER	FW Rules	Traffic leaks to ISP
2. NAT	NAT only on VPN_CHI + VPN_NYC	Outbound NAT	Unmasked traffic
3. DNS	Resolver outgoing = VPN only	DNS Resolver	DNS leak
4. IPv6	Blocked on all interfaces	FW Rules	IPv6 bypasses tunnel
5. DoH/DoT	DOH_IPS blocked on 443-853	FW Rules	Encrypted DNS bypass



Status / Gateways

Gateways | Gateway Groups

Gateways

Name	Gateway	Monitor	RTT	RTTsd	Loss	Status	Description	Action
WAN_DHCP (default)	[redacted]	[redacted]1	9.589ms	2.608ms	0.0%	Online	Interface WAN_DHCP Gateway	
GW_VPN_NYC	[redacted].105	9.9.9.9	60.706ms	2.002ms	0.0%	Online	Mullvad NYC Gateway	
GW_VPN_CHI	[redacted].85	1.1.1.1	40.522ms	2.253ms	0.0%	Online	Mullvad CHI Gateway	

Status / Gateways / Gateway Groups

Gateways | Gateway Groups

Gateway Groups

Group Name	Gateways		Description	Action
VPN_FAILOVER	Tier 1	Tier 2		
	GW_VPN_CHI Online			
		GW_VPN_NYC Online		

Status > Gateways + ipleak.net results

5. DHCP & DNS

Interface	Range	DNS	Gateway
LAN	10.10.1.10–245	10.10.1.1	10.10.1.1
VLAN10_USERS	10.10.10.10–245	10.10.10.1	10.10.10.1
VLAN20_IOT	10.10.20.10–254	10.10.20.1	10.10.20.1
VLAN30_GUEST	10.10.30.10–254	10.10.30.1	10.10.30.1
VLAN40_LAB	10.10.40.10–254	10.10.40.1	10.10.40.1
VLAN50_MGMT	10.10.50.10–254	10.10.50.1	10.10.50.1

5.1 Static Reservations

Device	IP	MAC	VLAN
Printer	10.10.20.5	[DHCP mapping]	20
R6400 AP	10.10.10.254	xx:xx:xx:xx:xx:xx	10
GS308E	10.10.1.100	xx:xx:xx:xx:xx:xx	1

5.2 DNS Resolver

Listens: LAN, all VLANs, Localhost, pfB DNSBL (10.10.255.1). Outgoing: VPN_CHI + VPN_NYC only. DNS Server Override: UNCHECKED.

⚠ **CAUTION** Never set Outgoing to WAN or All. DNS would leak to ISP.

6. Firewall Rules

Top-to-bottom, first match wins. Implicit deny at bottom. All VLANs (except MGMT) use VPN_FAILOVER + RFC1918 inter-VLAN block.

6.1 Aliases

Alias	Type	Values	Purpose
DOH_IPS	Hosts	Cloudflare, Google, Quad9, AdGuard DoH IPs	Block encrypted DNS
RFC1918	Networks	10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16	Block inter-VLAN
MSFT_CONNECTIVITY	Hosts	Microsoft NCSI IPs	Windows status (optional)

6.2 WAN

#	Action	Proto	Source	Dest	Port	GW	Description
1	Block	Any	RFC1918	*	*	none	Block private
2	Block	Any	Bogon	*	*	none	Block bogon
3	Pass	IPv4	*	pfB	*	none	pfBlockerNG

6.3 LAN

#	Action	Proto	Source	Dest	Port	GW	Description
1	Pass	Any	*	LAN addr	443,80	none	Anti-Lockout
2	Block	TCP/UDP	LAN sub	DOH_IPS	443-853	none	Block DoH/DoT
3	Block	TCP/UDP	LAN sub	*	53	WAN_DHCP	Block DNS to WAN
4	Pass	TCP	LAN sub	MSFT	443	WAN_DHCP	MS check (oE>>B4.6 ()N

#	Action	Proto	Source	Dest	Port	GW	Description
3	Pass	TCP	V10 sub	10.10.1.100	80-443	none	TEMP: Switch
4	Block	IPv4	V10 sub	RFC1918	*	none	Inter-VLAN block
5	Block	TCP/UDP	V10 sub	*	53	WAN_DHCP	DNS to WAN
6	Block	TCP/UDP	V10 sub	DOH_IPS	443-853	none	DoH/DoT
7	Pass	IPv4	V10 sub	*	*	FAILOVER	VPN
8	Block	IPv6	V10 sub	*	*	none	IPv6

⚠ **WARNING** TEMP rule #3 must be removed after switch config is done.

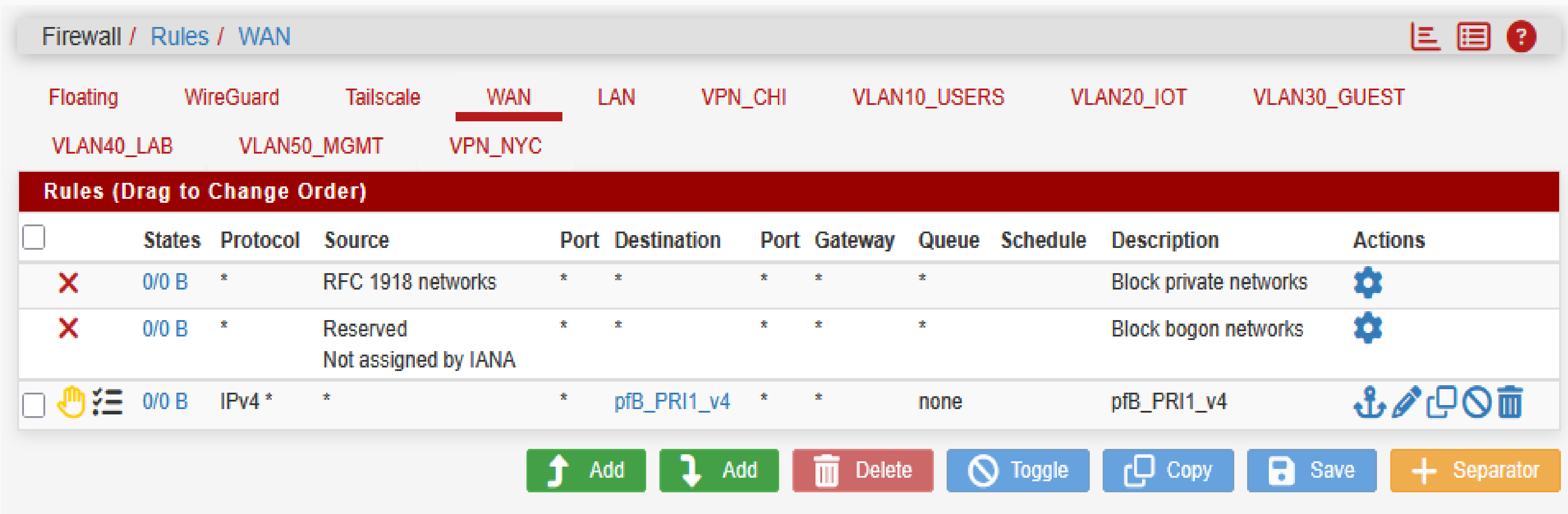
6.5 VLANs 20/30/40 (Same Pattern)

#	Action	Proto	Source	Dest	Port	GW	Description
1	Pass	IPv4	[VLAN] sub	[VLAN] addr	*	none	pfSense access
2	Block	TCP/UDP	[VLAN] sub	*	53	WAN_DHCP	DNS to WAN
3	Block	TCP/UDP	[VLAN] sub	DOH_IPS	443-853	none	DoH/DoT
4	Block	IPv4	[VLAN] sub	RFC1918	*	none	Inter-VLAN
5	Pass	IPv4	[VLAN] sub	*	*	FAILOVER	VPN
6	Block	IPv6	[VLAN] sub	*	*	none	IPv6

6.6 VLAN50_MGMT (Different)

#	Action	Proto	Source	Dest	Port	GW	Description
1	Pass	IPv4	V50 sub	V50 addr	*	none	pfSense access
2	Block	TCP/UDP	V50 sub	*	53	WAN_DHCP	DNS to WAN
3	Block	TCP/UDP	V50 sub	DOH_IPS	443-853	none	DoH/DoT
4	Pass	IPv4	V50 sub	*	*	none	Direct (no VPN)

⚠ **CAUTION** MGMT has no RFC1918 block (needs all-VLAN access) and no VPN (direct internet). Real ISP IP exposed.



Firewall > Rules for all interfaces

Floating | WireGuard | Tailscale | WAN | LAN | VPN_CHI | VLAN10_USERS | VLAN20_IOT | VLAN30_GUEST | VLAN40_LAB | VLAN50_MGMT | VPN_NYC

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	0/370 KiB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
✗	0/0 B	IPv4 TCP/UDP	LAN subnets	*	DOH_IPS	443 - 853	*	none			
✗	0/129 KiB	IPv4 TCP/UDP	LAN subnets	*	*	53 (DNS)	WAN_DHCP	none		Block DNS to WAN (anti-leak)	
✓	0/0 B	IPv4 TCP	LAN subnets	*	MSFT_CONNECTIVITY	443 (HTTPS)	WAN_DHCP	none			
✓	0/13 KiB	IPv4 *	LAN subnets	*	LAN subnets	*	*	none		Allow LAN-to-LAN (no VPN)	
✓	0/0 B	IPv4 *	LAN subnets	*	*	*	VPN_FAILOVER	none		Default allow LAN to any rule	
✗	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none			

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Floating | WireGuard | Tailscale | WAN | LAN | VPN_CHI | VLAN10_USERS | VLAN20_IOT | VLAN30_GUEST | VLAN40_LAB | VLAN50_MGMT | VPN_NYC

Rules (Drag to Change Order)

States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
3/24.63 MiB	IPv4 *	VLAN10_USERS subnets	*	VLAN10_USERS address	*	*	none		Allow access to pfSense gateway	
0/0 B	IPv4 TCP/UDP	VLAN10_USERS subnets	*	10.10.20.5	9100	*	none		Allow printing to VLAN20 printer	
0/0 B	IPv4 TCP	VLAN10_USERS subnets	*	10.10.1.100	80 - 443	*	none		TEMP_VLAN10_Switch Mgmt_(REMOVE AFTER)	
0/1.08 MiB	IPv4 *	VLAN10_USERS subnets	*	RFC1918	*	*	none		Block inter-VLAN — Isolate VLAN10	
0/40 KiB	IPv4 TCP/UDP	VLAN10_USERS subnets	*	*	53 (DNS)	WAN_DHCP	none		Block DNS to WAN — Anti-leak	
0/0 B	IPv4 TCP/UDP	VLAN10_USERS subnets	*	DOH_IPS	443 - 853	*	none		Block DoH/DoT — Prevent DNS bypas	
39/1.45 GiB	IPv4 *	VLAN10_USERS subnets	*	*	*	VPN_FAILOVER	none		VLAN10_to_Mullvad	
0/0 B	IPv6 *	VLAN10_USERS subnets	*	*	*	*	none		Block IPv6 — Prevent leaks	

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Firewall / Rules / VLAN20_IOT

Floating WireGuard Tailscale WAN LAN VPN_CHI VLAN10_USERS VLAN20_IOT VLAN30_GUEST VLAN40_LAB VLAN50_MGMT VPN_NYC

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
	0/0 B	IPv4 *	VLAN20_IOT subnets	*	VLAN20_IOT address	*	*	none		Allow access to pfSense gateway	
	0/1 KiB	IPv4 TCP/UDP	VLAN20_IOT subnets	*	*	53 (DNS)	WAN_DHCP	none		Block DNS to WAN — Anti-leak	
	0/0 B	IPv4 TCP/UDP	VLAN20_IOT subnets	*	DOH_IPS	443 - 853	*	none		Block DoH/DoT — Prevent DNS bypas	
	0/0 B	IPv4 *	VLAN20_IOT subnets	*	RFC1918	*	*	none		Block inter-VLAN — Isolate VLAN20	
	0/76 B	IPv4 *	VLAN20_IOT subnets	*	*	*	VPN_FAILOVER	none		VLAN20_to_Mullvad	
	0/0 B	IPv6 *	VLAN20_IOT subnets	*	*	*	*	none		Block IPv6 — Prevent leaks	

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Firewall / Rules / VLAN30_GUEST

Floating | WireGuard | Tailscale | WAN | LAN | VPN_CHI | VLAN10_USERS | VLAN20_IOT | VLAN30_GUEST | VLAN40_LAB | VLAN50_MGMT | VPN_NYC

Rules (Drag to Change Order)

		States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/0 B	IPv4 *	VLAN30_GUEST subnets	*	VLAN30_GUEST address	*	*	none		Allow access to pfSense gateway	
☐	✗ ⚙	0/0 B	IPv4 TCP/UDP	VLAN30_GUEST subnets	*	*	53 (DNS)	WAN_DHCP	none		Block DNS to WAN — Anti-leak	
☐	✗	0/0 B	IPv4 TCP/UDP	VLAN30_GUEST subnets	*	DOH_IPS	443 - 853	*	none		Block DoH/DoT — Prevent DNS bypas	
☐	✗	0/0 B	IPv4 *	VLAN30_GUEST subnets	*	RFC1918	*	*	none		Block inter-VLAN — Isolate VLAN30	
☐	✓ ⚙	0/0 B	IPv4 *	VLAN30_GUEST subnets	*	*	*	VPN_FAILOVER	none		VLAN30_to_Mullvad	
☐	✗	0/0 B	IPv6 *	VLAN30_GUEST subnets	*	*	*	*	none		Block IPv6 — Prevent leaks	

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Firewall / Rules / VLAN40_LAB

Floating WireGuard Tailscale WAN LAN VPN_CHI VLAN10_USERS VLAN20_IOT VLAN30_GUEST VLAN40_LAB VLAN50_MGMT VPN_NYC

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
	0/0 B	IPv4 *	VLAN40_LAB subnets	*	VLAN40_LAB address	*	*	none		Allow access to pfSense gateway	
	0/0 B	IPv4 TCP/UDP	VLAN40_LAB subnets	*	*	53 (DNS)	WAN_DHCP	none		Block DNS to WAN — Anti-leak	
	0/0 B	IPv4 TCP/UDP	VLAN40_LAB subnets	*	DOH_IPS	443 - 853	*	none		Block DoH/DoT — Prevent DNS bypas	
	0/0 B	IPv4 *	VLAN40_LAB subnets	*	RFC1918	*	*	none		Block inter-VLAN — Isolate VLAN40	
	0/0 B	IPv4 *	VLAN40_LAB subnets	*	*	*	VPN_FAILOVER	none		VLAN40_to_Mullvad	
	0/0 B	IPv6 *	VLAN40_LAB subnets	*	*	*	*	none		Block IPv6 — Prevent leaks	

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Firewall / Rules / VLAN50_MGMT

Floating WireGuard Tailscale WAN LAN VPN_CHI VLAN10_USERS VLAN20_IOT VLAN30_GUEST VLAN40_LAB VLAN50_MGMT VPN_NYC

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	0/0 B	IPv4 *	VLAN50_MGMT subnets	*	VLAN50_MGMT address	*	*	none		Allow access to pfSense gateway	
✕	0/0 B	IPv4 TCP/UDP	VLAN50_MGMT subnets	*	*	53 (DNS)	WAN_DHCP	none		Block DNS to WAN — Anti-leak	
✕	0/0 B	IPv4 TCP/UDP	VLAN50_MGMT subnets	*	DOH_IPS	443 - 853	*	none		Block DoH/DoT — Prevent DNS bypas	
✓	0/0 B	IPv4 *	VLAN50_MGMT subnets	*	*	*	*	none		VLAN50_MGMT — Direct internet (no VPN)	

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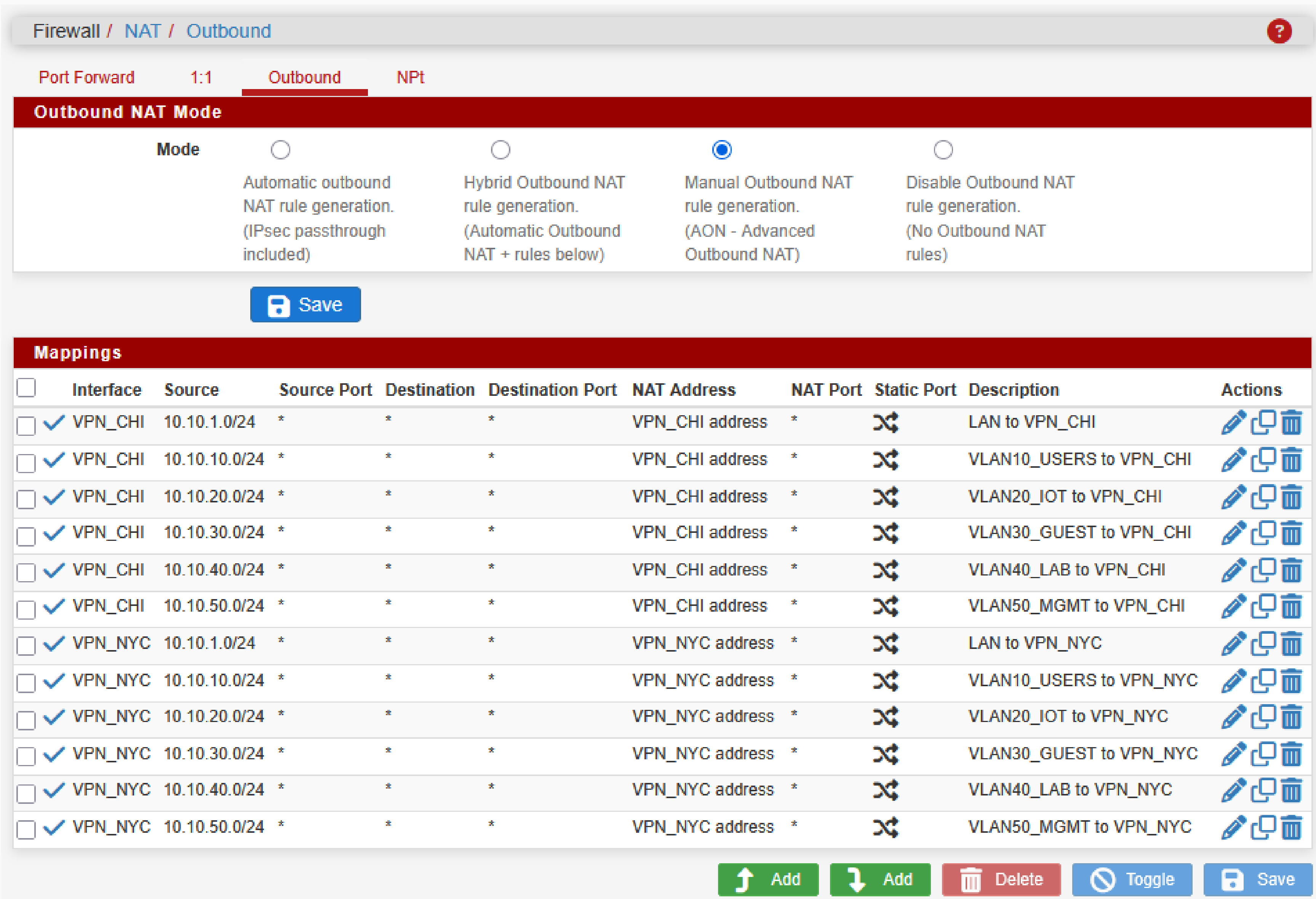
7. Outbound NAT

Manual mode. 12 rules: 6 per VPN interface. No WAN NAT for internal subnets = kill switch.

Interface	Source	NAT Address	Description
VPN_CHI	10.10.1.0/24	VPN_CHI addr	LAN→CHI
VPN_CHI	10.10.10.0/24	VPN_CHI addr	V10→CHI
VPN_CHI	10.10.20.0/24	VPN_CHI addr	V20→CHI
VPN_CHI	10.10.30.0/24	VPN_CHI addr	V30→CHI
VPN_CHI	10.10.40.0/24	VPN_CHI addr	V40→CHI
VPN_CHI	10.10.50.0/24	VPN_CHI addr	V50→CHI
VPN_NYC	10.10.1.0/24	VPN_NYC addr	LAN→NYC
VPN_NYC	10.10.10.0/24	VPN_NYC addr	V10→NYC
VPN_NYC	10.10.20.0/24	VPN_NYC addr	V20→NYC
VPN_NYC	10.10.30.0/24	VPN_NYC addr	V30→NYC
VPN_NYC	10.10.40.0/24	VPN_NYC addr	V40→NYC
VPN_NYC	10.10.50.0/24	VPN_NYC addr	V50→NYC



Firewall / NAT / Outbound

Port Forward | 1:1 | Outbound | NPt

Outbound NAT Mode

Mode

- ( ) Automatic outbound NAT rule generation. (IPsec passthrough included)
- ( ) Hybrid Outbound NAT rule generation. (Automatic Outbound NAT + rules below)
- (•) Manual Outbound NAT rule generation. (AON - Advanced Outbound NAT)
- ( ) Disable Outbound NAT rule generation. (No Outbound NAT rules)

Save

Mappings

Interface	Source	Source Port	Destination	Destination Port	NAT Address	NAT Port	Static Port	Description	Actions
VPN_CHI	10.10.1.0/24	*	*	*	VPN_CHI address	*		LAN to VPN_CHI	
VPN_CHI	10.10.10.0/24	*	*	*	VPN_CHI address	*		VLAN10_USERS to VPN_CHI	
VPN_CHI	10.10.20.0/24	*	*	*	VPN_CHI address	*		VLAN20_IOT to VPN_CHI	
VPN_CHI	10.10.30.0/24	*	*	*	VPN_CHI address	*		VLAN30_GUEST to VPN_CHI	
VPN_CHI	10.10.40.0/24	*	*	*	VPN_CHI address	*		VLAN40_LAB to VPN_CHI	
VPN_CHI	10.10.50.0/24	*	*	*	VPN_CHI address	*		VLAN50_MGMT to VPN_CHI	
VPN_NYC	10.10.1.0/24	*	*	*	VPN_NYC address	*		LAN to VPN_NYC	
VPN_NYC	10.10.10.0/24	*	*	*	VPN_NYC address	*		VLAN10_USERS to VPN_NYC	
VPN_NYC	10.10.20.0/24	*	*	*	VPN_NYC address	*		VLAN20_IOT to VPN_NYC	
VPN_NYC	10.10.30.0/24	*	*	*	VPN_NYC address	*		VLAN30_GUEST to VPN_NYC	
VPN_NYC	10.10.40.0/24	*	*	*	VPN_NYC address	*		VLAN40_LAB to VPN_NYC	
VPN_NYC	10.10.50.0/24	*	*	*	VPN_NYC address	*		VLAN50_MGMT to VPN_NYC	

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Firewall > NAT > Outbound

8. Switch (GS308E)

802.1Q Advanced. Port 1 trunk to pfSense. Ports 2-8 access.

8.1 Identity

Property	Value
Model	Netgear GS308E v4
MAC	xx:xx:xx:xx:xx:xx
Serial	xxxxxxxxxxxx
Firmware	xx.xx.xxxx
IP	10.10.1.100
Gateway	10.10.1.1
Factory IP	192.xxx.0.xxx

8.2 Port Assignments

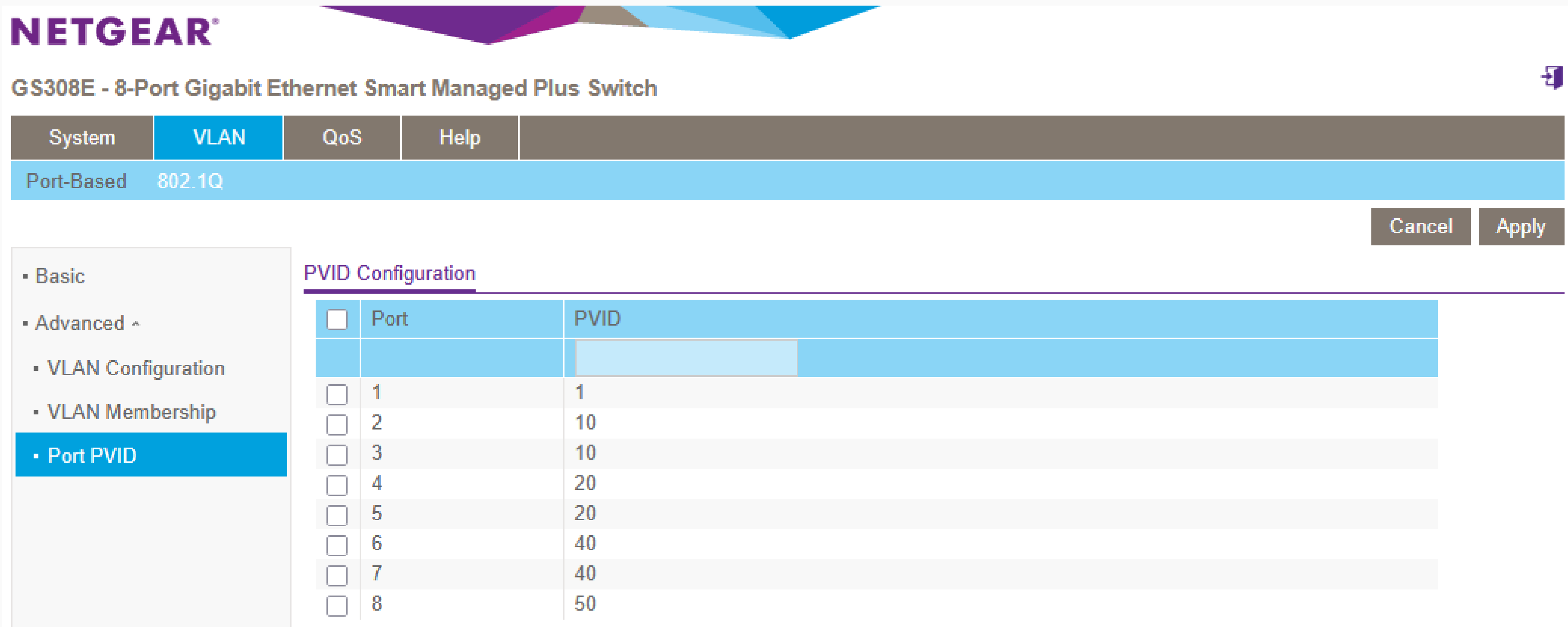
Port	Device	VLAN	Mode	PVID	Notes
1	pfSense igb1	All	Trunk	1	T:10,20,30,40,50 U:1
2	PC	10	Access	10	Workstation
3	R6400 AP	10	Access	10	Wi-Fi bridge
4	Printer	20	Access	20	Static 10.10.20.5
5	IoT	20	Access	20	Reserved
6	Lab	40	Access	40	Cisco 3560
7	Lab	40	Access	40	Cisco 1900
8	MGMT	50	Access	50	Break-glass

8.3 Membership Matrix

VLAN	P1	P2	P3	P4	P5	P6	P7	P8
1	**U**	—	—	—	—	—	—	—
10	**T**	**U**	**U**	—	—	—	—	—
20	**T**	—	—	**U**	**U**	—	—	—
30	**T**	—	—	—	—	—	—	—

VLAN	P1	P2	P3	P4	P5	P6	P7	P8
40	**T**	—	—	—	—	**U**	**U**	—
50	**T**	—	—	—	—	—	—	**U**

HINT T=Tagged (blue), U=Untagged (green), —=Not member

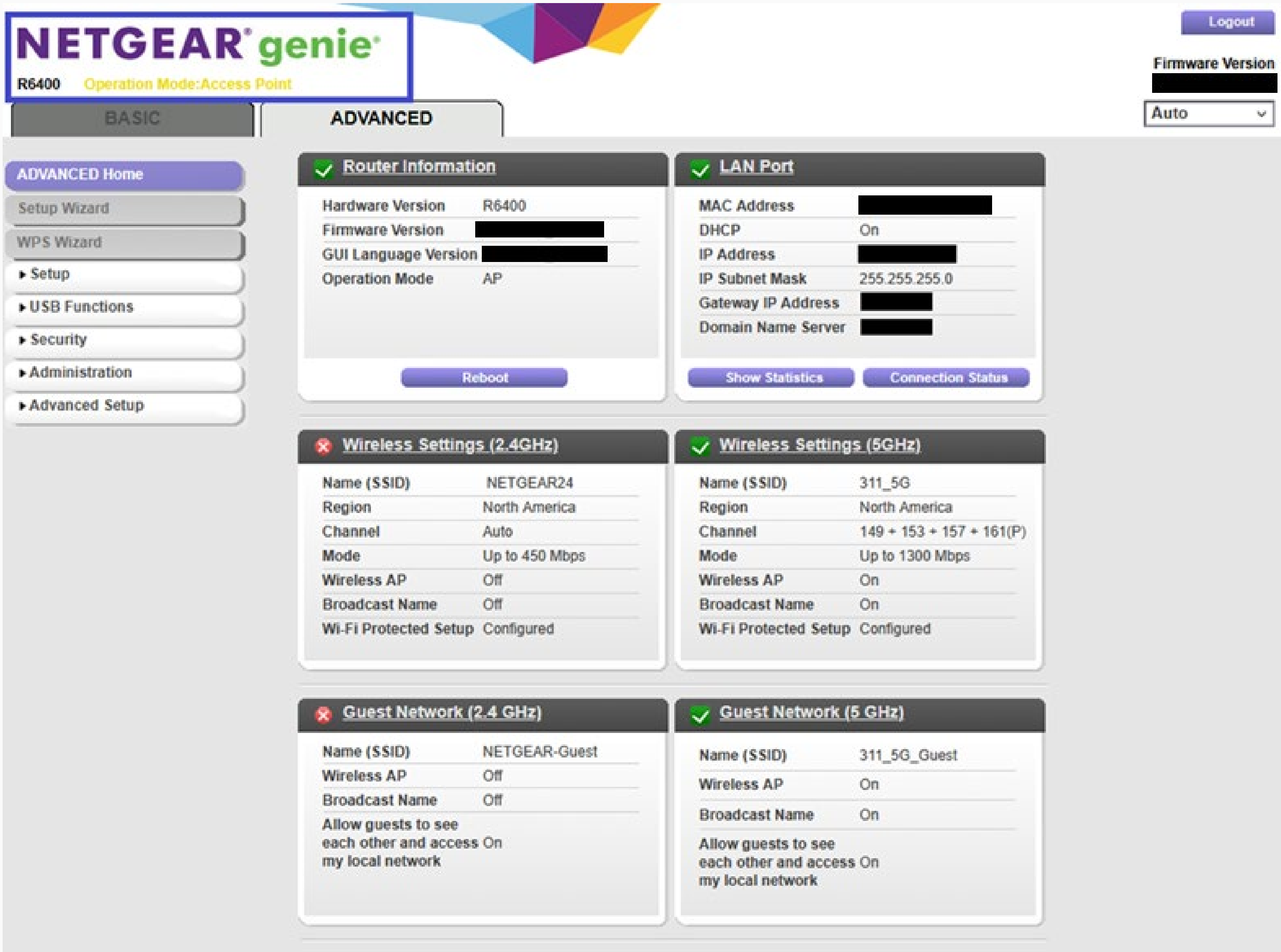


GS308E VLAN Config + Membership + PVID pages

9. Access Point (R6400)

Setting	Value
Device	Netgear R6400 (AC1750)
Mode	AP only (router/NAT/DHCP off)
IP	10.10.10.254 (fixed)
MAC	xx:xx:xx:xx:xx:xx
Port	GS308E Port 3 (VLAN 10)
Wi-Fi	Dual-band 2.4+5 GHz

INFO All Wi-Fi clients land on VLAN 10. Traffic routes through VPN. AP is a pure Layer 2 bridge.



R6400 AP mode config page

10. IP Address Allocation (CMDB)

IP	Device	MAC	VLAN	Type
10.10.1.1	pfSense LAN GW	—	1	Static
10.10.1.100	GS308E Switch	xx:xx:xx:xx:xx:xx	1	Static
10.10.10.1	pfSense VLAN10 GW	—	10	Static
10.10.10.254	R6400 AP	xx:xx:xx:xx:xx:xx	10	Fixed
10.10.20.1	pfSense VLAN20 GW	—	20	Static
10.10.20.5	Printer	[DHCP]	20	Reserved
10.10.30.1	pfSense VLAN30 GW	—	30	Static
10.10.40.1	pfSense VLAN40 GW	—	40	Static
10.10.50.1	pfSense VLAN50 GW	—	50	Static
10.10.255.1	pfBlockerNG DNSBL	—	VIP	Virtual

⚠ **CAUTION** 10.10.255.1 is pfBlockerNG Virtual IP. DO NOT DELETE.

11. Troubleshooting

11.1 No Internet

#	Check	How	Expected
1	VPN gateways	Status > Gateways	CHI+NYC online
2	WAN	Status > Gateways	WAN_DHCP online
3	Ping	Diag > Ping > 1.1.1.1	Replies
4	DNS	Diag > DNS Lookup	Resolves
5	NAT	FW > NAT > Outbound	12 VPN rules, 0 WAN

11.2 Lost Switch Access

1. Check TEMP rule on VLAN10. 2. Ping 10.10.1.100 from pfSense. 3. Emergency: PC=192.168.0.100/24, browse 192.168.0.239.

11.3 Lost pfSense GUI

VLAN10: https://10.10.10.1. Direct to igb1: https://10.10.1.1 (static PC 10.10.1.200/24).

12. Backup & Restore

🛑 **BACKUP** ALWAYS backup before ANY change. Format: pfsense-YYYY-MM-DD-before-[change].xml

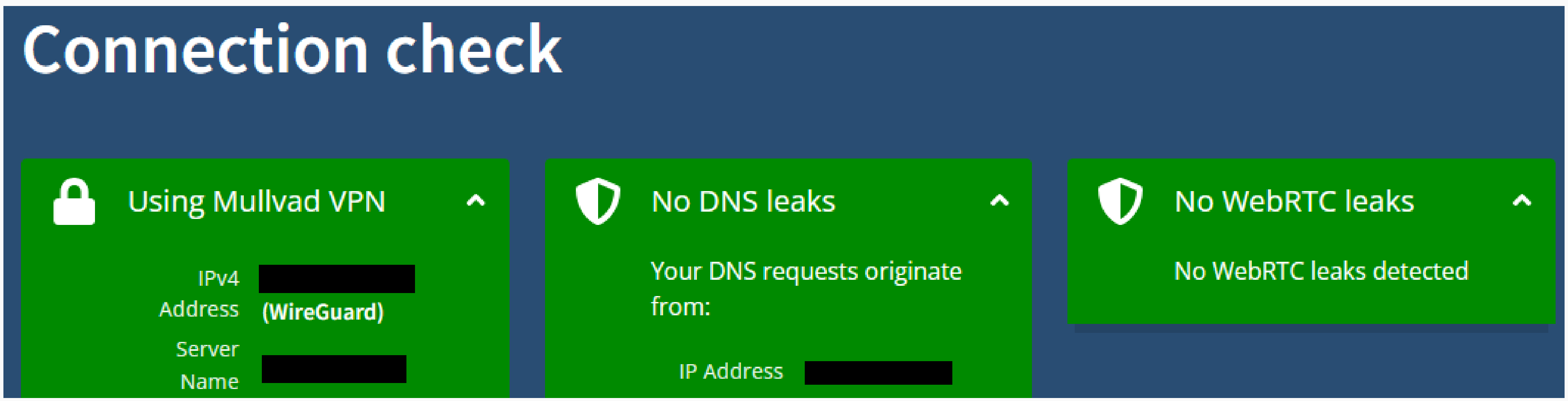
Device	How	Naming
pfSense	Diagnostics > Backup > Download XML	pfsense-YYYY-MM-DD-desc.xml
Switch	System > Maintenance > Export	gs308e-YYYY-MM-DD.cfg
AP	Administration > Backup	r6400-YYYY-MM-DD.cfg

12.1 Backups Taken

File	When
pfsense-2026-03-06-before-phase2.xml	Before Phase 2
pfsense-2026-03-06-before-lan-renumber.xml	Before LAN change
pfsense-2026-03-06-before-vlan10-renumber.xml	Before VLAN10 change
pfsense-2026-03-06-after-killswitch-audit.xml	After 5 fixes
pfsense-2026-03-08-final.xml	After all Phase 2

13. Testing Results (March 2,8,11…., 2026)

Test	Method	Result	Status
VPN active	ipleak.net	Mullvad IP shown	✅ PASS
DNS leak	mullvad.net/check	No leaks	✅ PASS
WebRTC	mullvad.net/check	No leaks	✅ PASS
Inter-VLAN	ping 10.10.20/30/40/50.1	All blocked	✅ PASS
Gateway	ping 10.10.10.1	0% loss	✅ PASS
Failover	Disabled CHI	NYC took over	✅ PASS
Kill switch	Both tunnels down	Zero internet	✅ PASS



ipleak.net + mullvad.net/check results

14. Future Planning

#	Change	Impact	Status and Comment
1	Remove TEMP switch rule	Low	Conditional (now off)
2	Guest Wi-Fi SSID for VLAN30	Medium	Hardware limitation (can't do that in R6400 Router (AP Mode)
3	Printer ports 443/631	Low	As needed (I think Port 9100 RAW is good for my lab)
4	Tailscale remote access	Medium	Planned (In process)
5	IPv6 block on VLAN50	Low	Optional (Done)
6	Guest rate limiting	Medium	Future
7	IDS/IPS (Suricata)	High	Future
8	West Coast VPN endpoint	Medium	Future

All Status and Comment last update: 03-10-2026 11:11 AM

14.1 Change Procedure

1. Document change. 2. Backup. 3. Implement. 4. Test. 5. Leak test. 6. Update doc. 7. Post-change backup.

References

Protectli FW6E: https://protectli.com/wp-content/uploads/2025/02/FW6E-Datasheet-20250204.pdf

R6400 Manual: https://www.downloads.netgear.com/files/GDC/R6400/R6400_UM_07Aug2015.pdf

GS308E Manual: https://www.downloads.netgear.com/files/GDC/GS308Ev4/GS308Ev4_UM_EN.pdf

pfSense Docs: https://docs.netgate.com/pfsense/en/latest/